

ENGR 3324: Signals and Systems
Sample Test 3

Time:

Name:

Q1) Find the zero-state response $y_{zs}(t)$, if the input is $x(t) = 4e^{-3t}u(t)$ to the system described by the differential equation.

$$\frac{d^2y(t)}{dt^2} + 9y(t) = \frac{dx(t)}{dt} + x(t)$$

Q2) Find the zero-state response $y_{zs}(t)$, if the input is $x(t) = 4e^{-3t}u(t)$ and the system transfer function is

$$H(s) = \frac{s+3}{s^2+6s+8}$$

Q3) Find the steady state response (zero-state response), if the input is $x(t) = 4\cos(5t)$ and the system transfer function is

$$H(s) = \frac{s+3}{s^2+6s+8}$$

Q4) Draw the block diagram of Form II realization (canonic) for the system specified by the transfer function

$$H(s) = \frac{3s+2}{s^2+6s+9}$$

Q5) Design a third order filter that has a magnitude frequency response shown below. Show the locations of the poles and zeros on the s-plane and find the transfer function $H(s)$.

